

Question	Answer	Marks
1(a)	force $\times$ <u>perpendicular</u> distance (of line of action of force to a point) or product of force and perpendicular distance (to a point)	B1
1(b)(i)	in (rotational) equilibrium	B1
	<u>sum / total</u> of CW moments about a point = <u>sum / total</u> of ACW moments about the (same) point.	B1
1(b)(ii)	component of weight = $75 \times \cos 42^\circ$ = 56 N	A1
1(b)(iii)	$F \times 80$ or 56×40 or $F \times 0.8$ or 56×0.4 $F \times 80 = 56 \times 40$	C1
	$F = 28$ N	A1

Question	Answer	Marks
2(a)	product of mass and velocity	B1
2(b)	change in momentum = $(-1.4) - (+2.8)$ = $(-) 4.2 \text{ kg m s}^{-1}$	A1
2(c)	$F = \Delta p / (\Delta)t$ or $F = \text{gradient}$ = $4.2 / 12$	C1
	= 0.35 N	A1
2(d)	constant / uniform (rate of) decrease (of speed to zero).	B1